1 Quantitative exclusion of arithmetic translation algebras

Keep the unchanged odd-dimensional operator $U_N = \exp(i\tau N A_N) M_N$, with $\tau = 10^{-80}$, and the physical midpoint symbol, including its shifted momentum coefficient. Put $\kappa = \tau/(2\pi)$ and equip matrices with $\langle X, Y\rangle_N = N^{-1}\operatorname{Tr}(X^*Y)$. For an odd divisor $d \geq 3$ of N, let

$$E^0_{d,N} = \operatorname{span}\{T_N((N/d)v) : v \in (\mathbb{Z}/d\mathbb{Z})^2 \setminus \{0\}\}, \qquad \Pi^0_{d,N} : \mathcal{H}_N \longrightarrow E^0_{d,N}.$$

The $d^2 - 1$ displayed raw Weyl modes are exactly orthonormal. No coprimality of d and N/d or CRT factorization is assumed.

Theorem 1 (Arithmetic translation-algebra exclusion)**.** *There is C_τ, independent of N, d and the grid labels, such that*

$$\left\|\Pi^0_{d,N} \operatorname{Ad}(U_N)\Pi^0_{d,N}\right\| \leq \min\left\{1, C_\tau \left(\frac{d^{262}}{N}\right)^{1/129}\right\}.$$

For every fixed $\delta < 1/262$, uniformly when $d \mid N$ and $d \leq N^\delta$, the compression is

$$O_\tau\big(N^{-(1-262\delta)/129}\big).$$

In particular, for $d \leq N^{1/524}$,

$$\sup_{\substack{X \in E^0_{d,N} \\ \|X\|_N = 1}} \left|\|U_N X U_N^* - X\|_N^2 - 2\right| \leq 2C_\tau N^{-1/258}.$$

The fixed-d statement follows immediately. Since the cat propagator permutes these labels, the conclusion applies to normalized arithmetic cat orbit sums. If a projection P_N lies in $\mathbb{C}I \oplus E^0_{d,N}$ and $0 < \alpha_N = \operatorname{rank}(P_N)/N < 1$, its nonzero centered part $X_N = (P_N - \alpha_N I)/\sqrt{\alpha_N(1-\alpha_N)}$ satisfies the same unit-vector estimate. The projections $0, I$ are excluded from this normalization. Arbitrary spectral projections are not covered.

Uniform finite type at fixed coupling. Write the kick flow as $\Psi = \Phi_g^{-\kappa} = \mathrm{Id} + u$. Complex-strip bounds give $u = -\kappa X_g + R$. Crucially, for a translation ξ and $r = \|\xi\|_\infty$ in its centered representative,

$$\|D^\alpha(R(\cdot + \xi) - R)\|_\infty \leq C\kappa^2 r \qquad (|\alpha| \leq 16),$$

using one extra derivative. Degree-16 differential filters isolate the primitive $(-1, 0)$ and $(0,-1)$ Fourier modes of the actual symbol. Their coefficient norms are $6409325/1143072 < 6$, and their surviving multipliers have moduli $4/3$ and 2. Explicit Cauchy estimates bound the filtered remainder by $10^{55}\kappa^2 r$, smaller than the leading $|\kappa| r$ signal at the literal coupling. A selected real component of $\xi - \eta + u(x + \xi) - u(x)$ has finite type at most 16, uniformly in the translations. Directional slicing and the one-dimensional sublevel lemma therefore give coincidence-sublevel area $O_\tau((d\varepsilon)^{1/16})$ for a nonzero d-grid translation.

Quantum overlap and dimension loss. Exact Weyl multiplication gives the actual kick Egorov error $O_\tau(N^{-2}\|a\|_{H^5})$. It remains uniform after inserting a translation, since translations preserve Sobolev norms. In the trace intertwining argument, division by squared map separation costs ε^{-4}

in C^3. Hilbert–Schmidt Cauchy–Schwarz on the bad set retains the *square root* of its area, while the aliased Fourier tail costs $N^{-2}\varepsilon^{-3}$. Thus every matrix overlap is bounded by

$$C_\tau \left((d\varepsilon)^{1/32} + N^{-1}\varepsilon^{-4} + N^{-2}\varepsilon^{-3} \right).$$

Choosing $\varepsilon = N^{-32/129} d^{-1/129}$ gives $C_\tau (d^4/N)^{1/129}$ per entry. Passing to the entire (d^2-1)-dimensional span costs a further factor d^2, yielding the stated $262/129$ exponent. The residual estimate follows from unitarity and $\| \mathrm{Ad}(U_N)X - X \|_N^2 = 2 - 2\,\mathrm{Re}\langle X, \mathrm{Ad}(U_N)X \rangle_N$.

Scope. This is a project proof with independent internal review and exact finite controls, not an external referee certificate or analytic formalization. The exponents are not claimed optimal; the endpoint $\delta = 1/262$ is not a decay statement. Constants depend on the fixed coupling, with no useful numerical startup or uniform shrinking-coupling estimate supplied. Faster-growing divisors, general quantum commutants, coherent returns and the signed Heisenberg joint-trace comparison remain open. The theorem does not establish microscopic BGS universality.